🏗️ M002 Extension 3 — Build Your Own 6-Maze Challenge

Topic: Design a Maze + Write the Solution · **Course:** Maze Madness · **Level:** Extension (Advanced) · **Time:** about 75 minutes

This is the big **builder challenge** for students who have finished the weekly mazes. This time you are not solving someone else's maze — **you are the maze maker.** You will design one long maze that **stretches through 6 different sections**, wire it with pistons and doors, and then write the **solution code** that drives the agent all the way to the end.

Open the **M002 EXT 3 — Cube World** and build inside it. You drive the agent with **chat commands** just like before: `l` turns left, `r` turns right, `run` starts your solver, and `rl` teleports the agent back to you.

Your maze must include at least:

- **2 pistons** (each opens a wall or path when powered)
- **2 doors** (the agent opens them with `agent.interact(...)`)
- **4 AND conditions** in your solution code (`... and ...`)
- **1 OR condition** in your solution code (`... or ...`)

Keep this page beside you while you build — you will check off each part as you go.

1 · Plan Your 6 Mazes 🗺️

A great maze is **planned before it is built.** Your maze runs through **6 sections** in a row — the agent finishes one and walks straight into the next. Give each section one job.

Sketch your 6 sections. For each one, write the section number and what makes it tricky (a turn, a piston, a door, a junction where the agent must check two things).

Where will your 2 pistons go? Which sections, and what does each piston open?

Where will your 2 doors go? What is behind each one?

2 · Build the Maze 🧱

Now build it in the world. Walk the path yourself first (no code) and make sure a person can get from start to end. **Check off each part as you place it.**

> ☐ Section 1 built
>
> ☐ Section 2 built
>
> ☐ Section 3 built
>
> ☐ Section 4 built
>
> ☐ Section 5 built
>
> ☐ Section 6 built
>
> ☐ Piston #1 placed and tested
>
> ☐ Piston #2 placed and tested
>
> ☐ Door #1 placed
>
> ☐ Door #2 placed
>
> ☐ A clear START block and a clear GOAL block

Did anything break while you built it? What did you have to change?

3 · Write the Solution 💻

Now write the **solver** — the code that gets the agent through your whole maze. Use a `while True` loop so it checks on every step.

Here is the shape to start from. Fill it in for **your** maze. Remember the rule: your solution needs **at least 4 AND conditions and 1 OR condition.**

```
while True:
    agent.move(FORWARD, 1)

    # a junction — turn only when BOTH things are true (AND)
    if agent.detect(BLOCK, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)

    # a door OR a piston gate — handle either one (OR)
    elif agent.detect(BLOCK, AHEAD) or agent.detect(REDSTONE, RIGHT):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

Write your full solution here. Add more `if` / `elif` checks until it covers all 6 sections. Mark each AND with `# AND` and your OR with `# OR` in a comment so you can count them.

4 · Count Your Logic ✅

Go back through the solution you just wrote and **count.** Write the number, then copy out one real line from your code as proof.

How many AND conditions did you use? (need at least 4)

Copy one of your AND lines here:

How many OR conditions did you use? (need at least 1)

Copy your OR line here:

Why did that one spot need an OR instead of an AND? What two different things is it checking for?

5 · Test It and Finish 📸

Type `run` and watch the agent. It will get stuck somewhere — that is normal. Find where it stops, fix that part of your code or your maze, and run it again. Keep going until the agent reaches the **GOAL** all by itself. Use `r1` to send it back to the start between tries.

When the agent finishes your maze on its own, come back here. Send a photo or video of the agent at the goal, then explain what you did. Use these sentence starters — write 5 to 7 sentences total.

> My maze runs through 6 sections. The trickiest section was ...
>
> I put my 2 pistons in ... because ...
>
> The agent opens my doors with `agent.interact(...)`, which works because ...
>
> I used an AND condition at ... to make the agent ...
>
> I needed an OR condition at ... because ...
>
> The agent kept getting stuck at ... until I ...
>
> If I built a version 2, I would add ...

6 · Present Your Maze 🎥

You built this maze, so you are the expert. Take a video on your phone (or a parent's phone) and **present your maze like a designer showing off their game.** Try to use these words: **section**, **piston**, **door**, **detect**, **AND**, **OR**, **loop**.

> 1. Show the whole maze from the start and point out the 6 sections.
> 2. Show each piston and each door, and say what it does.
> 3. Read one AND check from your code out loud and say what decision it makes.
> 4. Read your OR check and say why it needed OR.
> 5. Type `run` and let the agent solve your maze to the goal.

Write what you will say in your video. Plan it here first — you can read from it while filming.

Submit ✅

Send this worksheet + your walkthrough video to teacher on KakaoTalk.